

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0201

Ritzville 138 kV
September 20, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0201. The Interconnection Customer proposes a 112.6 MW solar photovoltaic facility interconnecting at the Ritzville 138 kV substation in the Maryland area. The request was received on February 02, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 135 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,281,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Artesia Country Club / Air Force Base 345 kV, the G-P Gypsum / Brighton 138 kV loads to 159.9% of its 207 MVA rating. Project DFAX: 6.4%.

For the loss of Montgomery Village / Bondurant 138 kV, the Montgomery Village / Bondurant Substation 138 kV #2 loads to 155.6% of its 261 MVA rating. Project DFAX: 18.7%.

The Montgomery Village / Bondurant 138 kV #1 is overloaded to 152.3% of its 267 MVA emergency rating for the loss of the Montgomery Village / Bondurant 138 kV. The project contributes 18.7% DFAX (21.1 MW) to this overload.

The Bondurant / Tramway 138 kV is overloaded to 150.3% of its 259 MVA emergency rating for the loss of the Bondurant / Tramway 138 kV Ckt 2. The project contributes 18.6% DFAX (21.0 MW) to this overload.

Bondurant / Tramway 138 kV #2: 150.3% loading (259 MVA) following the outage of the Tramway / Bondurant 138 kV; DFAX = 18.6%.

For the loss of Taber / Dixleta 345 kV, the Brighton / Drum 138 kV loads to 140.9% of its 211 MVA rating. Project DFAX: 5.0%.

G-P Gypsum / Ross Fork 138 kV: 117.0% loading (215 MVA) following the outage of the Clover Vly / G-P Gypsum 138 kV L; DFAX = 6.3%.

The Tramway / Scarcello 138 kV is overloaded to 112.5% of its 226 MVA emergency rating for the loss of the Rossmoor S / Tramway 138 kV L. The project contributes 15.4% DFAX (17.3 MW) to this overload.

The Radfords Run Wind Farm / McCormick 138 kV is overloaded to 111.9% of its 274 MVA emergency rating for the loss of the Radfords Run Wind Farm / Haddam Neck Substation 138 kV L. The project contributes 5.9% DFAX (6.7 MW) to this overload.

The Rancho / Middlebrook 138 kV L is overloaded to 106.3% of its 214 MVA emergency rating for the loss of the Air Force Base / Artesia Country Club 345 kV Line. The project contributes 10.9% DFAX (12.3 MW) to this overload.

Rockville / Phillips Pump 1 138 kV L: 103.1% loading (161 MVA) following the outage of the Tramway / Rossmoor S Substation 138 kV; DFAX = 15.4%.

Air Force Base / Rancho 138 kV line: 102.7% loading (199 MVA) following the outage of the Artesia Country Club / Air Force Base 345 kV; DFAX = 10.9%.

The St Francois Substation / Radfords Run Wind Farm 138 kV is overloaded to 100.9% of its 663 MVA emergency rating for the loss of the Boulder Basin / Baseline Substation 138 kV line. The project contributes 9.1% DFAX (10.3 MW) to this overload.

The Hollydale Substation / St Francois 138 kV is overloaded to 100.6% of its 679 MVA emergency rating for the loss of the Baseline / Boulder Basin 138 kV. The project contributes 9.1% DFAX (10.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
G-P Gypsum / Brighton 138 kV	Rebuild line with higher-capacity conductor	\$7,081,000	42% (\$2,973,000)
Montgomery Village / Bondurant Substation 138 kV #2	Rebuild line with higher-capacity conductor	\$6,697,000	53% (\$3,576,000)
Montgomery Village / Bondurant 138 kV #1	Rebuild line with higher-capacity conductor	\$8,709,000	53% (\$4,651,000)
Bondurant / Tramway 138 kV	Rebuild line with higher-capacity conductor	\$2,930,000	53% (\$1,565,000)
Bondurant / Tramway 138 kV #2	Rebuild line with higher-capacity conductor	\$1,646,000	53% (\$879,000)
Brighton / Drum 138 kV	Rebuild line with higher-capacity conductor	\$9,416,000	100% (\$9,416,000)
G-P Gypsum / Ross Fork 138 kV	Rebuild line with higher-capacity conductor	\$6,774,000	100% (\$6,774,000)
Tramway / Scarcello 138 kV	Rebuild line with higher-capacity conductor	\$842,000	40% (\$338,000)
Radfords Run Wind Farm / McCormick 138 kV	Rebuild line with higher-capacity conductor	\$7,611,000	2% (\$155,000)
Rancho / Middlebrook 138 kV	Reconductor line	\$798,000	59% (\$467,000)
Rockville / Phillips Pump 1 138 kV L	Reconductor line	\$451,000	40% (\$181,000)
Air Force Base / Rancho 138 kV line	Reconductor line	\$475,000	59% (\$278,000)
St Francois Substation / Radfords Run Wind Farm 138 kV	Reconductor line	\$1,083,000	2% (\$18,000)
Hollydale / St Francois Substation 138 kV	Reconductor line	\$619,000	2% (\$10,000)
	Total allocated to project		\$31,281,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.